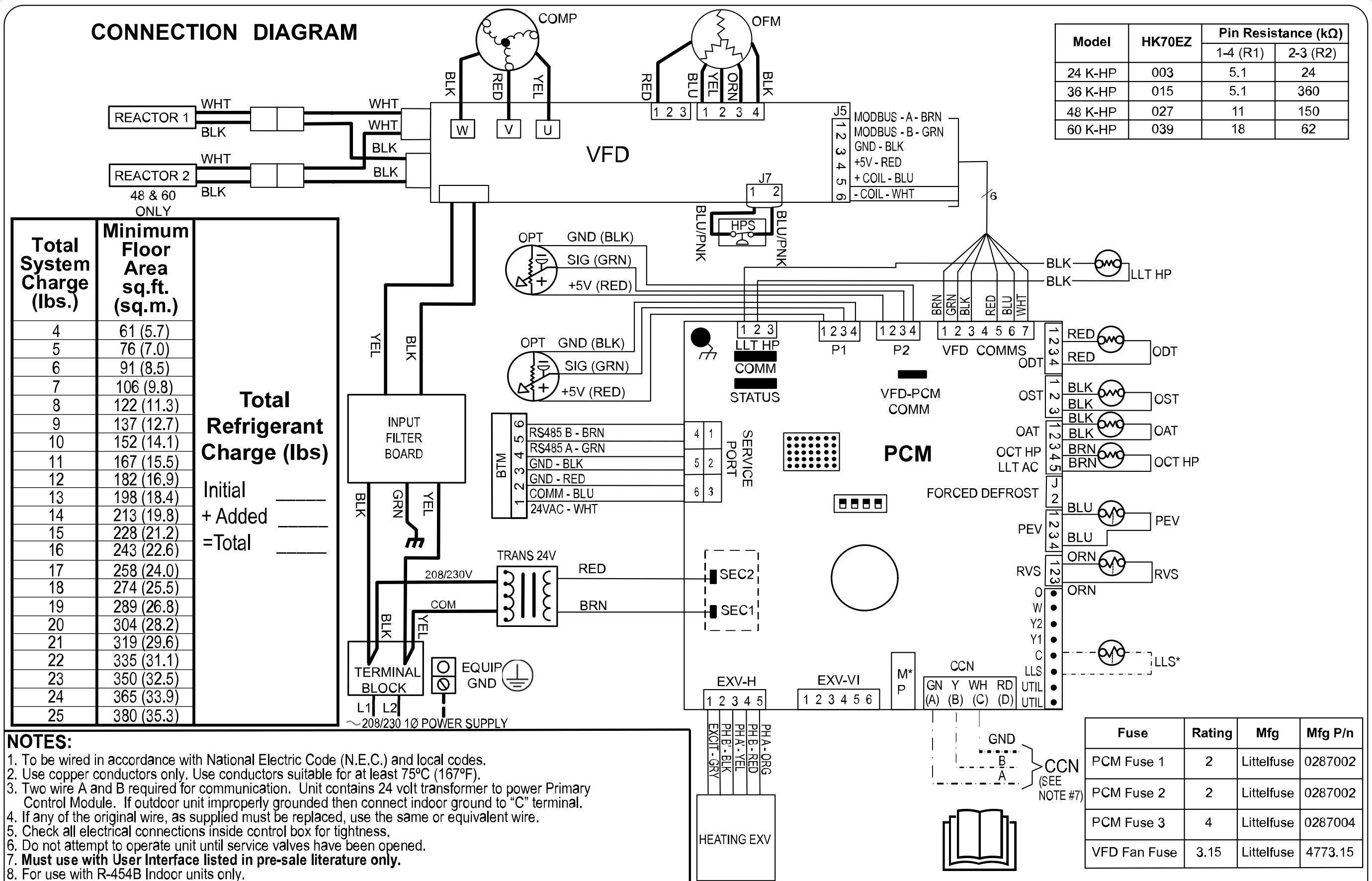
27VNA0 Infinity™
Variable Speed Heat Pump
with Puron Advance™ Refrigerant
2 to 5 Tons

Wiring Diagram



Model	HK70EZ	Pin Resistance (kΩ)	
		1-4 (R1)	2-3 (R2)
24 K-HP	003	5.1	24
36 K-HP	015	5.1	360
48 K-HP	027	11	150
60 K-HP	039	18	62

Total System Charge (lbs.)	Minimum Floor Area sq.ft. (sq.m.)
4	61 (5.7)
5	76 (7.0)
6	91 (8.5)
7	106 (9.8)
8	122 (11.3)
9	137 (12.7)
10	152 (14.1)
11	167 (15.5)
12	182 (16.9)
13	198 (18.4)
14	213 (19.8)
15	228 (21.2)
16	243 (22.6)
17	258 (24.0)
18	274 (25.5)
19	289 (26.8)
20	304 (28.2)
21	319 (29.6)
22	335 (31.1)
23	350 (32.5)
24	365 (33.9)
25	380 (35.3)

Total Refrigerant Charge (lbs)

Initial _____
+ Added _____
=Total _____

Fuse	Rating	Mfg	Mfg P/n
PCM Fuse 1	2	Littelfuse	0287002
PCM Fuse 2	2	Littelfuse	0287002
PCM Fuse 3	4	Littelfuse	0287004
VFD Fan Fuse	3.15	Littelfuse	4773.15

NOTES:

1. To be wired in accordance with National Electric Code (N.E.C.) and local codes.
2. Use copper conductors only. Use conductors suitable for at least 75ºC (167ºF).
3. Two wire A and B required for communication. Unit contains 24 volt transformer to power Primary Control Module. If outdoor unit improperly grounded then connect indoor ground to "C" terminal.
4. If any of the original wire, as supplied must be replaced, use the same or equivalent wire.
5. Check all electrical connections inside control box for tightness.
6. Do not attempt to operate unit until service valves have been opened.
7. **Must use with User Interface listed in pre-sale literature only.**
8. For use with R-454B Indoor units only.
9. Factory authorized dissipation control board must be Installed with Indoor unit.

UNIT OPERATION

This Control Board Contains A Five Minute Short Cycle Protector. A Five Minute Delay Will Occur Between Compressor Off/on Cycles. To Bypass Delay, Short Forced Defrost Pins For 1 Second Then Release.

- **FIELD INITIATED FORCED DEFROST** - (Shown As Forced Defrost On Board) By Placing A Jumper Across The Forced Defrost Terminals For 5 Seconds, Or Longer, And Then Removing The Jumper The Unit Will Initiate A Defrost Cycle Regardless Of Coil Temperature. The Defrost Cycle Will Terminate At 65ºF (+/-5ºF) If Coil Temperature Is Above 32ºF Or Outdoor Ambient Temperature Is Above 50ºF, Defrost Mode Will Terminate After 30 Seconds Of Active Mode.
- **OUTDOOR PRESSURE TRANSDUCERS** - The P1 And P2 On The PCM Are Independent Of Where The OPTs Are Installed On The Tubing. Both OPTs Can Be Connected To Either P1 Or P2.

-LEGEND-

FACTORY POWER WIRING
FIELD POWER WIRING
FACTORY CONTROL WIRING
FIELD CONTROL WIRING
BTM BLUETOOTH MODULE
COMM CCN COMMUNICATION LIGHT
COMP COMPRESSOR
EXV ELECTRONIC EXPANSION VALVE
HPS HIGH PRESSURE SWITCH
LLS* LIQUID LINE SOLENOID
LLT HP LIQUID LINE TEMPERATURE THERMISTOR
MP* MODEL PLUG
OAT OUTDOOR AIR TEMPERATURE THERMISTOR
OCT HP OUTDOOR COIL TEMPERATURE THERMISTOR
ODT OUTDOOR DISCHARGE TEMPERATURE THERMISTOR
OFM OUTDOOR FAN MOTOR
OPT OUTDOOR PRESSURE TRANSDUCER
OST OUTDOOR SUCTION TEMPERATURE THERMISTOR
PCM PRIMARY CONTROL MODULE (AC/HP CONTROL BOARD)
PEV PRESSURE EQUALIZATION VALVE
RVS REVERSING VALVE
STATUS SYSTEM FUNCTION LIGHT
UTIL UTILITY CURTAILMENT
VFD VARIABLE FREQUENCY DRIVE

* MAY NOT BE FACTORY INSTALLED

350388-101 REV.B

Fig. 1 – Size 2 - 5 Tons

Edition Date: 09/25

Catalog No: 27VNA0-1W

Replaces: New